

MRI Image Quality Metrics (IQMs)

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December 9, 2025

Abstract

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

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Introduction

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

1 Per-Slice IQMs (Foreground / Background pixels)

All metrics below accept foreground pixels $\mathbf{F}$ (from segmentation) and background pixels $\mathbf{B}$. NaN/Inf inputs yield NaN. For BG-dependent metrics (SNR1, SNR2, SNR4, CNR, CJV, FBER), if background is missing or unusable (empty, zero variance, non-finite), we fall back to foreground homogeneity μ_F/σ_F to avoid artificial NaNs.

1.1 Mean (MEAN)

Foreground mean intensity:

$$\text{MEAN} = \mu_F = \frac{1}{N_F} \sum_i F_i$$

1.2 Range (RNG)

Foreground intensity range:

$$\text{RNG} = \max(F) - \min(F)$$

1.3 Variance (VAR)

Foreground variance:

$$\text{VAR} = \sigma_F^2 = \frac{1}{N_F} \sum_i (F_i - \mu_F)^2$$

1.4 Coefficient of Variation (CV)

Foreground coefficient of variation (percent):

$$\text{CV} = \frac{\sigma_F}{\mu_F} \times 100$$

1.5 Contrast per Pixel (CPP)

Laplacian-filtered contrast, mean response of a 3×3 Laplacian:

$$f_1 = \frac{1}{8} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}, \quad \text{CPP} = \text{mean}(f_1 * F)$$

where $*$ is 2D convolution (foreground is median-filtered only when used for PSNR below).

1.6 Peak Signal-to-Noise Ratio (PSNR)

$$\text{PSNR} = 10 \log_{10} \left(\frac{\max(F)^2}{\text{MSE}(F, \hat{F})} \right)$$

where $\hat{F}$ is either the background $\mathbf{B}$ (if provided) or a median-filtered version of $\mathbf{F}$.

1.7 SNR1

Foreground/background std; fallback to foreground stability if background unusable [1], [2]:

$$\text{SNR1} = \begin{cases} \frac{\sigma_F}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise (fg homogeneity)} \end{cases}$$

1.8 SNR2

Foreground mean over background std; fallback to foreground stability if background unusable:

$$\text{SNR2} = \begin{cases} \frac{\mu_F}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.9 SNR3

Foreground standard deviation divided by centered-foreground standard deviation (stability check):

$$\text{SNR3} = \frac{\sigma_F}{\sigma_{F-\mu_F}}$$

1.10 SNR4

Foreground mean over background mean; fallback to foreground stability if background unusable:

$$\text{SNR4} = \begin{cases} \frac{\mu_F}{\mu_B}, & \mu_B \text{ finite and } \mu_B \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.11 Contrast-to-Noise Ratio (CNR)

Foreground/background mean difference over background std; fallback to foreground stability if background unusable:

$$\text{CNR} = \begin{cases} \frac{\mu_F - \mu_B}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.12 Patch Coefficient of Variation (CVP)

Foreground patch coefficient of variation:

$$\text{CVP} = \frac{\sigma_F}{\mu_F}$$

1.13 Coefficient of Joint Variation (CJV)

Joint variation between foreground and background; fallback to foreground stability if background unusable:

$$\text{CJV} = \begin{cases} \frac{\sigma_F + \sigma_B}{|\mu_F - \mu_B|}, & \text{finite and } |\mu_F - \mu_B| \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.14 Entropy Focus Criterion (EFC)

Normalized Shannon entropy of foreground:

$$p_i = \frac{F_i}{\sqrt{\sum_j F_j^2}}, \quad \text{EFC} = -\frac{1}{\log N_F} \sum_i p_i \log p_i$$

higher values suggest more ghosting/noise.

1.15 Foreground-Background Energy Ratio (FBER)

Median energy ratio; fallback to foreground stability if background unusable:

$$\text{FBER} = \begin{cases} \frac{\text{median}(F^2)}{\text{median}(B^2)}, & \text{finite and median}(B^2) \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.16 Skewness (SKW)

Foreground skewness (third standardized moment):

$$\text{SKW} = \frac{\frac{1}{N_F} \sum_i (F_i - \mu_F)^3}{\sigma_F^3}$$

1.17 Excess Kurtosis (KURT)

Foreground excess kurtosis (fourth standardized moment minus 3):

$$\text{KURT} = \frac{\frac{1}{N_F} \sum_i (F_i - \mu_F)^4}{\sigma_F^4} - 3$$

2 Participant-Level QC

2.1 Failure Count (FAIL_FRAC)

Counts total non-finite IQM evaluations (NaN/Inf) across all slices/metrics for a participant. Higher counts flag poor FG/BG segmentation or corrupted slices:

$$\text{FAIL_FRAC} = \sum_{s \in \text{slices}} \sum_{m \in \text{IQMs}} \mathbf{1}\{\text{IQM}_{m,s} \text{ is non-finite}\}$$

References

- [1] A. R. Sadri, A. Janowczyk, R. Zhou, *et al.*, “Mrqy—an open-source tool for quality control of mr imaging data,” *Medical physics*, vol. 47, no. 12, pp. 6029–6038, 2020.
- [2] R. Kushol, A. H. Wilman, S. Kalra, and Y.-H. Yang, “Dsmri: Domain shift analyzer for multi-center mri datasets,” *Diagnostics*, vol. 13, no. 18, p. 2947, 2023.